Kinematic and Dynamic Analysis of a 4-Actuator Driving Simulator Platform

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2025-05-14

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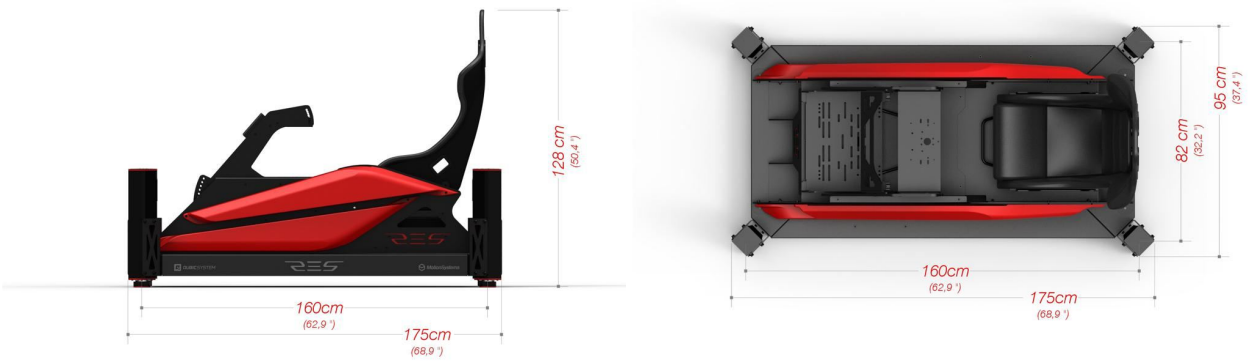
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Objective

The objective of the homework is to study the **kinematic behavior** and **workspace** of a motion platform controlled by four vertical linear actuators, and to develop a **dynamic model** that describes the platform's response to different actuator inputs. As optional extensions, students can implement with Simscape the model of the linear electrical actuators coupled with a basic **feedback control system** and simulate the closed-loop response of the platform. Bonus point will be given for a correct implementation of this optional task.

Project Description

Students will analyze a simplified **3-degree-of-freedom (DOF)** platform which is used in the driving simulator of the Department of Industrial Engineering of University of Trento. The platform is mounted on four electrical **vertical linear actuators**, which control its **heave (Z)**, **pitch (rotation about X)**, **roll (rotation about Y)**.



The platform is considered rigid and connected at four base points (actuator bases on the floor) and four moving points on the platform (actuator tops) via prismatic joints. The linear actuators are orthogonal to the platform main base. See the picture below.

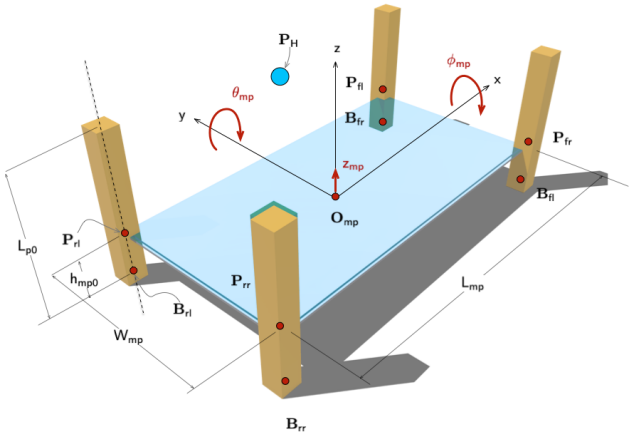


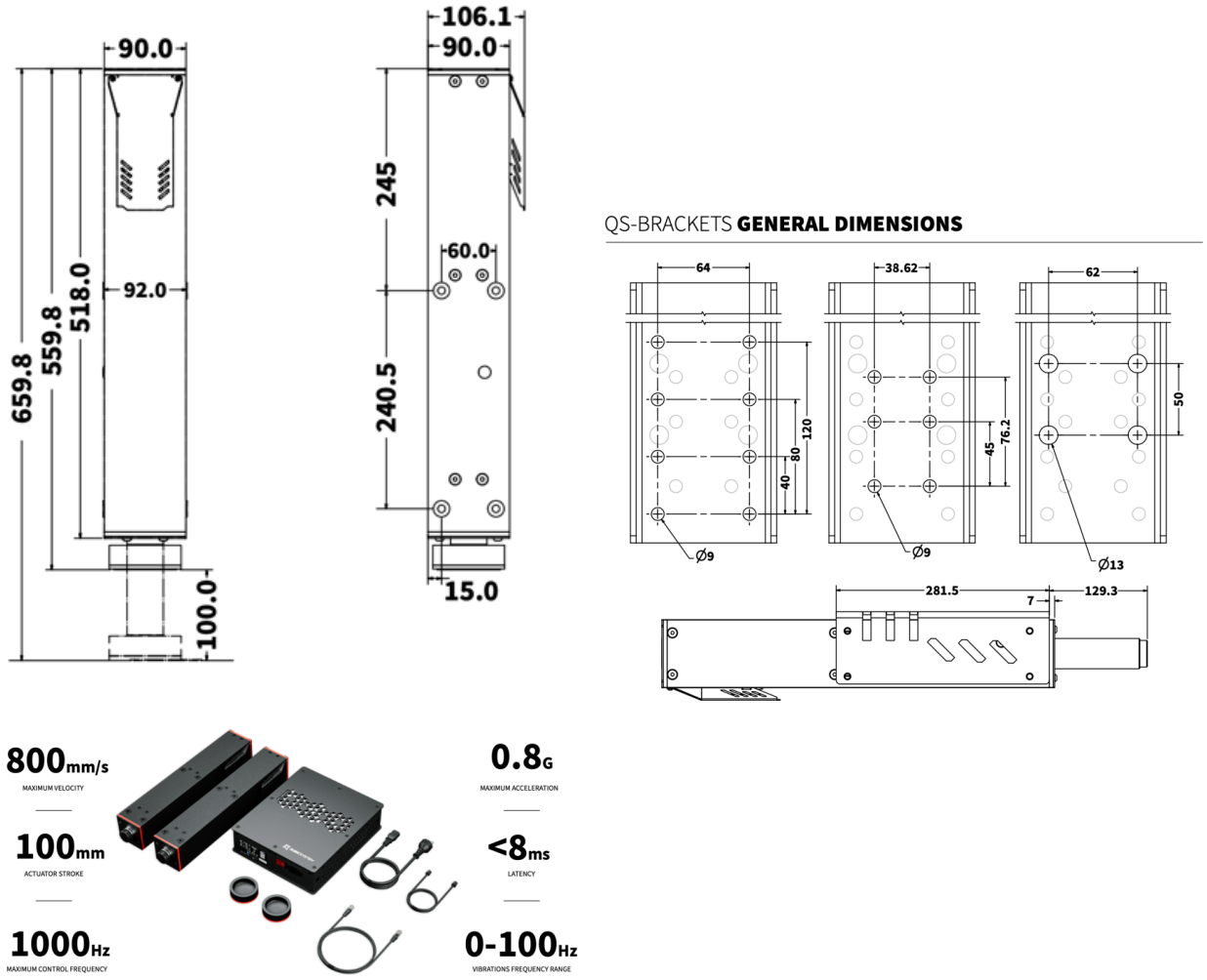
Figure 1: Schematic the driving simulator platform

The main base hosts a steat with a steering wheel and pedals. The platform is designed to simulate the motion of a car, allowing the driver to experience acceleration, braking, and cornering forces. The platform is controlled by a computer system that receives input from the driver which in combination with a vehicle dynamic model generates the appropriate actuator commands to simulate the desired motion. The actuators are capable of providing high-speed and high-precision movements vertical motion that can be mapped into **heave (Z)**, **pitch (rotation about X)**, **roll (rotation about Y)**. Thus the platform is not able to simulate lateral motion (translation along X and Y axes) or yaw (rotation about Z axis) but should be able to provide a realistic experience of the vehicle dynamics in terms of vertical accelerations and pitch/roll angles.

The objective is to derive the forward and inverse kinematics of the system, analyze the workspace of the platform, and develop a dynamic model that describes the platform's response to different actuator inputs.

1. System Geometry and Setup

The base layout, platform layout, actuator lengths, and attachment point coordinates are shown in the figures below.



MOTION PLATFORM **QS-220**

The main body of the platform is considered a rigid body with known connection points to actuators as shown in the figures.

More information can be found at these links: actuators and motion platform. Not all the inertial and geometrical parameters are available in the datasheet. You can estimate the value of the missing parameters making some assumptions that you have to justify.

The main geometrial parameters are in the table below (the meaning of some parameters can be found in the figure Figure 1):

Parameter	Value	Units
Platform max. payload	290	kg
Platform mass	150	kg

Parameter	Value	Units
Actautor mass	30	kg
Actuator stroke	100	mm
Actuator max. vel	800	mm/s
Actuator max. acc	0.8	g
Actuator height	0.5598	m
W___mp	0.82	m
L___mp	1.6	m
h___mp0	0.1896	m

The toal mass and inertia of the system is a combination of the subcomponents: platform, actuators , and driver. You have also to add the mass of the driver. You can assume that the driver is a rigid body with a mass of 75 kg.

The actuators are assumed to be ideal linear actuators with no friction or backlash. However, in the dynamic model of the electro-mechanic linear actuator you can create a more detailed model which may include also friction. This is for the optinal task.

Homework Tasks

The taks of the homework are divided into three main sections: **kinematic analysis**, **dynamic modeling**, and **system response analysis**. Each section builds upon the previous one, leading to a comprehensive understanding of the platform's behavior.

1. Kinematic Analysis

To complete the this task you need to derive the kinematic model using the transformation matrix and the Linear Graph approach. The kinematic model should be implemented in MATLAB using the Symbolic Math Toolbox and the provided `MultiBodyLibrary`. You can also use Maple as an alternative and `MBSymba` for the symbolic derivation of the equations of motion. The expected model is a 3D model of the platform with its 6 dof and the 4 actuators lengths as input. With the kinematic model you have to study the following points

- Study the **forward kinematics**: determine platform position and orientation based on actuator lengths.
- Derive the **inverse kinematics**: for a given platform pose (position and orientation), compute the required actuator lengths.
- Given **constraints** and **limits** of actuator motion indetify the motion ranges for heave, roll and picth
- Define a point that rapresent the position of the driver head and compute the velocity and acceleration perceived by the driver in that point in the body fixed frame.
- Calculate the **workspace envelope** of the platform by evaluating all reachable poses given actuator limits.
- Visualize the workspace in 3D using MATLAB, or similar tools.
- Discuss the implications of the workspace on the design and operation of the driving simulator.

2. Dynamic Modeling

To complete this task you need to derive the dynamic model of the system using the Newton-Euler or Lagrangian approach. The dynamic model should be implemented in MATLAB using the provided `MultiBodyLibrary`. You can also use Maple as an alternative and `MBSymba` for the symbolic derivation of the equations of motion. The expected model is a 3D model of the platform with its 6 dof and the 4 actuators lengths as input. With the dynamic model you have to study the following points

- Derive the **equations of motion** for the platform using Newton-Euler or Lagrangian methods.
- The linear actuator lengths can be used directly as inputs or you may model the actuator dynamics using a firt order dynamics.
- Convert the model for numerical solution in Matlab or in Maple
- Create an equivalent model in Simscape

3. System Response Analysis

- Apply different input signals to the actuators (e.g., sinusoidal, step inputs) and analyze:
  - Platform displacement
  - Angular response (pitch, roll)
  - Dynamic coupling between axes
  - compare the result with SimScape model for the same inputs Discuss the results obtained and what are the limitatiosn that you can see the paltform has and envisage how you can use the model for design purposes.

4. Optional Extensions

This section is optional and will give you bonus points of well done. The goal is to implement a more detailed model of the electrical actuator and a basic feedback control system using SimScape. The optional task is to be done after the completion of the previous tasks ano not doing this task will not affect the evaluation of the oevrall hoemwrok. - Implement the **dynamic model of the electrical actuator** using SimScape electrical library. The model should include: - Electrical circuit (voltage, current, resistance) - Mechanical dynamics (force, velocity, position) - Friction and backlash (if applicable) - Actuator limits (stroke, speed, acceleration) have a look at the QubicSystem datasheet for more information about power and current consumption - Implement a basic **feedback control system** (e.g., PID control) to achieve desired platform positions. - Add sensor models (encoders, IMUs) and simulate a closed-loop control scenario.

Deliverables

The final deliverables for the project should include: - **MATLAB Code** or **Maple Code** for (zip file that contains all the code): - Kinematic model - Dynamic model - Simulation scripts - SimScape model (if applicable)

- **Technical Report** detailing:

  - Kinematic derivations
  - Workspace analysis
  - Dynamic modeling approach
  - Simulation results
  - Discussion of limitations and potential improvements

Learning Outcomes

By completing this project, students will be able to:

- Apply kinematic principles to mechatronic systems using the Linear Graph approach
- Derive and analyze the dynamic behavior of a motion platform
- Develop and simulate real-world mechanical systems
- Understand the interaction between mechanical design and control